Page 1: Introduction to Artificial Intelligence & Machine Learning

What is Artificial Intelligence?

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn. The core goal of AI is to perform tasks such as visual perception, speech recognition, decision-making, and language translation.

Types of Machine Learning

Machine Learning (ML) is a subfield of AI focused on building systems that learn from data. There are three primary paradigms: Supervised Learning, Unsupervised Learning, and Reinforcement Learning. In Supervised Learning, models train on labeled datasets containing inputs and known outputs. Examples include Linear Regression, Logistic Regression, and Support Vector Machines (SVM). In Unsupervised Learning, algorithms discover hidden patterns or clusters in unlabeled data, such as K-Means clustering and Principal Component Analysis (PCA). In Reinforcement Learning, an agent learns to make optimal decisions by interacting with an environment through trial-and-error, guided by rewards and penalties.

Page 2: Deep Learning & Neural Network Architectures

Deep Learning Fundamentals

Deep Learning is a subset of machine learning based on artificial neural networks with multiple hidden layers. Inspired by the human brain, deep neural networks can automatically learn hierarchical representations of complex data such as images, audio, and raw text.

Activation Functions and Backpropagation

Neural networks introduce non-linearity through activation functions. The Rectified Linear Unit (ReLU) is widely used because it accelerates convergence and mitigates the vanishing gradient problem. Sigmoid and Tanh functions squish activations between 0 to 1 and -1 to 1 respectively. Training neural networks relies on Backpropagation, an efficient algorithm that computes the gradient of the loss function with respect to each network weight using the chain rule of calculus. Optimization algorithms like Stochastic Gradient Descent (SGD) and Adam iteratively update the weights to minimize loss.

Page 3: Natural Language Processing & Information Retrieval

Natural Language Processing (NLP)

Natural Language Processing (NLP) enables computers to understand, interpret, and manipulate human language. Key preprocessing steps include tokenization, lowercase normalization, stop-word removal, and lemmatization.

TF-IDF and Cosine Similarity in Document Search

TF-IDF stands for Term Frequency-Inverse Document Frequency. It evaluates the importance of a word within a document relative to a collection of documents (corpus). Term Frequency measures how frequently a word occurs in a specific passage. Inverse Document Frequency scales down common words (like 'the', 'is') and elevates informative terms. Cosine Similarity calculates the cosine of the angle between two multi-dimensional vectors. When a user asks a question, both the question and document passages are transformed into TF-IDF vectors. The cosine similarity metric produces a score between 0.0 and 1.0, identifying the most relevant passage independent of passage length.

Page 4: Model Evaluation and Overfitting Prevention

Evaluation Metrics for Classification

Accuracy measures the proportion of correct predictions out of all predictions. However, when classes are imbalanced, accuracy can be misleading. Precision calculates the proportion of true positives among all predicted positives, answering: of all predicted positives, how many were truly positive? Recall measures the ability of a model to identify all relevant instances, answering: of all actual positives, how many did the model catch? The F1 Score provides the harmonic mean of precision and recall, balancing both metrics.

Overfitting and Regularization Techniques

Overfitting occurs when a machine learning model memorizes training noise and fails to generalize to unseen test data. Standard techniques to prevent overfitting include K-Fold Cross-Validation, L1 (Lasso) and L2 (Ridge) Regularization, Dropout in neural networks, and Early Stopping during model training.